

#30 Podcast Transcription, Show Notes and Auto-Publishing

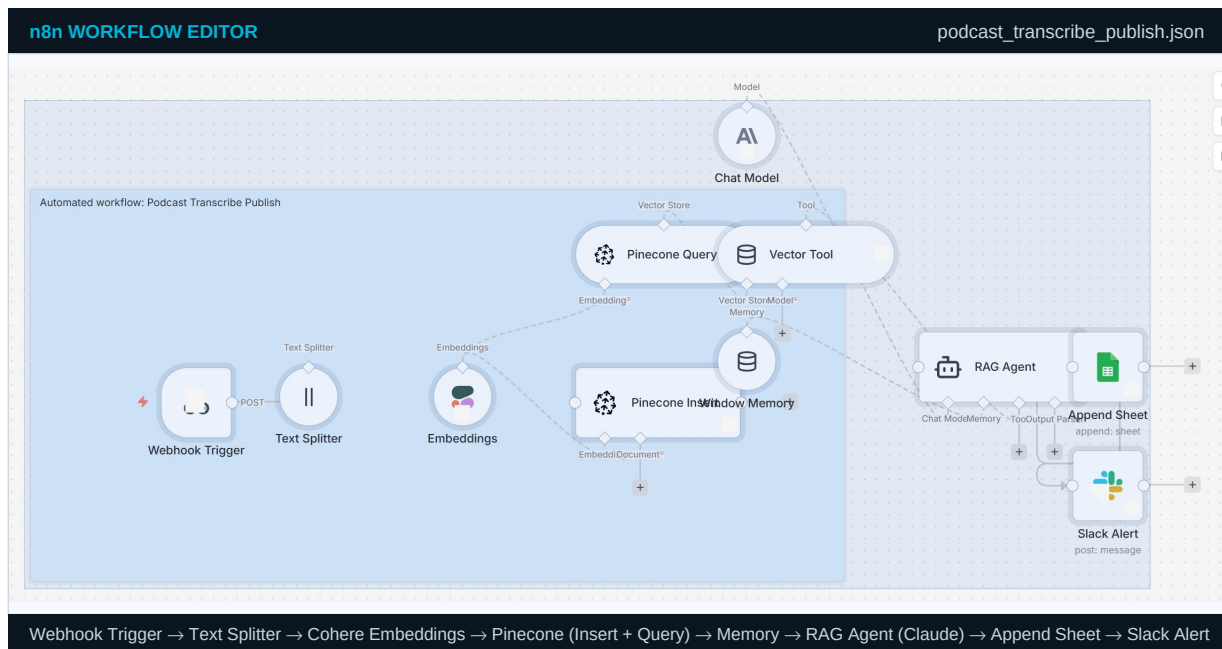
Media / AI
Podcast
Production

Upload audio to full content suite — published in 15 minutes automatically

BUSINESS PROBLEM

Podcast post-production takes 4-6 hours per episode: transcription, show notes, SEO blog post, social clips, and RSS update. This workflow processes raw audio, runs Whisper transcription, generates all content with Claude, and publishes to WordPress and social channels — 15 minutes total.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — audio file uploaded to Google Drive or S3

```

{
  "file_url": "https://drive.google.com/file/d/1Bxi.../download",
  "episode_number": 47,
  "episode_title": "Agentic AI in 2024 — What Actually Works",
  "guest_name": "Dr. Rania Al-Sabah",
  "duration_min": 45,
  "language": "en",
  "podcast_slug": "aiops-weekly"
}
  
```

OUTPUT — Publishing manifest — all content generated and published

```

{
  "json": {
    "episode": 47,
    "transcript_words": 8400,
    "whisper_ver_pct": 3.1,
    "show_notes_words": 460,
    "chapter_count": 12,
    "blog_url": "https://podcast.example.com/ep47-agentic-ai-2024",
    "linkedin_post_id": "7208341234567892",
    "rss_updated": true,
    "spotify_live": true,
    "processing_sec": 822
  }
}
  
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — audio file upload event	Google Drive webhook or S3 event fires: {file_url, episode_number, title, guest_name, duration, language}.
2	✂ Splitter — transcript chunking	Post-Whisper transcript split into 1,000-char chunks at natural speech pause boundaries.
3	■ Cohere Embeddings — transcript vectors	Each chunk embedded. Enables semantic search for best quote extraction and chapter detection.
4	■ Pinecone Insert — episode namespace	Full transcript stored with namespace: {podcast_slug}/{episode_number}. All episodes searchable.
5	■ Pinecone Query — key moments	Queries: 'find most quotable insight', 'find topic transitions for chapters', 'find key takeaways'.
6	■ Vector Tool — cross-episode search	Agent queries past episodes: 'has guest discussed this topic before?' Enables cross-episode references.
7	■ Window Buffer Memory — content context	Tracks episode topic, guest bio, key themes for consistent tone across all deliverables.
8	■ Claude — content generation suite	Sequentially: Whisper transcription → show notes → SEO blog → LinkedIn post → Twitter thread → chapter markers.
9	■ RAG Agent — orchestrates publishing	Calls: WordPress REST API for blog, Drive for transcript, LinkedIn API, RSS feed update.
10	■ Append Sheet — episode tracker	episode, word_count, blog_url, social_ids, publish_status, processing_time.
11	■ Slack — production complete	'Episode 47 published in 13m 42s' with links to blog, transcript, and all social posts.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI Whisper API	platform.openai.com — \$0.006 per minute of audio	OpenAI API
Cohere API Key	dashboard.cohere.com → API Keys	Cohere API
Pinecone API Key	app.pinecone.io → Create index: 1,024 dims, cosine	Pinecone API
Anthropic (Claude) API Key	console.anthropic.com → API Keys	Anthropic API
WordPress Application Password	WP Admin → Users → Profile → Application Passwords → Add New	WordPress API
Slack Bot Token	api.slack.com/apps → OAuth & Permissions	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario

An AI podcast publishes weekly 45-minute episodes. Host was spending 5-6 hours per episode on post-production: transcription, show notes, social posts, RSS, and WordPress.

Input received	Episode 47 uploaded at 14:00. File: 87MB, 45-minute interview with Dr. Rania Al-Sabah on agentic AI.
What the workflow found	Whisper transcribed 8,400 words in 3 minutes 12 seconds (WER 3.1%). Pinecone found 12 natural chapter breaks. Cross-episode search: guest was also on episode 31 — reference added to show notes.
Output produced	All deliverables in 13 minutes 42 seconds: transcript saved, show notes with 12 chapters, blog post published, LinkedIn and Twitter queued. Host review time: 8 minutes. API cost: \$0.27.

JSON: podcast_transcribe_publish.json	GitHub: trm-podcast-production-agent	openai.com/research/whisper
---------------------------------------	--------------------------------------	-----------------------------